Variance Reduction and Importance Sampling

Mark M. Fredrickson (mfredric@umich.edu)

Module 1: Integrate

Computational Methods in Statistics and Data Science (DATASCI 406)

The Cost of Precision

Recap: Monte Carlo answers are estimates

Last video: to compute $\theta = \mathrm{E}\left(g(X)\right)$ (or any integral we can write that way), we **sample and average**:

$$\hat{\theta}_n = \frac{1}{n}\sum_{i=1}^{n} g(X_i)$$

The CLT gave a confidence interval, and big-O gave the error rate:

$$\text{error} \approx \tau\, n^{-1/2} = O(n^{-1/2}), \quad \tau^2 = \mathrm{Var}\left(g(X)\right)$$

n sets the **rate**; τ sets the **constant**. The rate is fixed — every estimator today is still a sample mean.

But τ is not: different sampling schemes can estimate the **same** θ **with different** τ.

Today: **decrease the constant**.

Picking n to achieve a width

After we find a way to estimate, we can pick n to **achieve given precision**.

The CI width will be approximately:

$$w = 2z_{\alpha/2}\tau/\sqrt{n} \Rightarrow n = 4z_{\alpha/2}^2\tau^2/w^2$$

where $z_{\alpha/2} = P(Z \leq \alpha/2)$, $Z \sim N(0,1)$ and $\tau^2 = \mathrm{Var}\left(g(X)\right)$.

We need to **make a good guess** for τ^2.

- Estimate using a small sample.
- Find an upper bound and use the worst case.

Example: Finding an upper bound

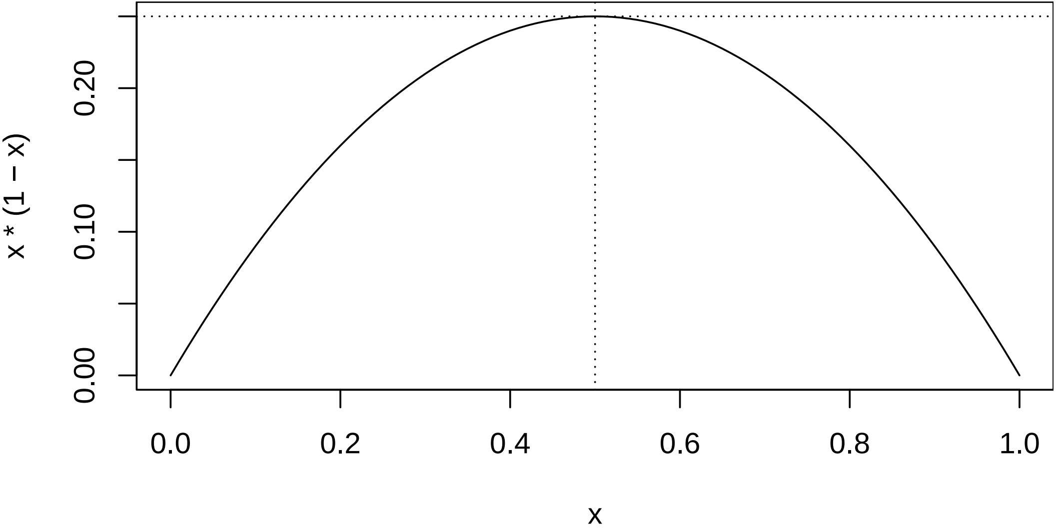
Let's estimate a probability for $X \sim N(0,1)$ again:

$$\theta = P(X \leq 1.96)$$

Here, $X_i \sim N(0,1)$, $g(x) = \mathbf{1}(x \leq 1.96)$. Recall,

$$g(X_i) \sim \text{Bernoulli}(\theta) \Rightarrow \text{Var}(g(X_i)) = \theta(1-\theta)$$

Where is the max of $\theta(1-\theta)$?



95% CI with $w = 0.001$

```
(targetN <- 4 * qnorm(0.975)^2 * 0.25 / 0.001^2)
```

```
## [1] 3841459
```

```
gxs <- rnorm(targetN) <= 1.96
(ci <- as.numeric(t.test(gxs)$conf.int))    ## 95% is the default
```

```
## [1] 0.9749 0.9752
```

```
diff(ci)
```

```
## [1] 0.0003121
```

Width is well less than our target of 0.001

Often we can find **two different estimators of the same** θ using RVs X and Y:

In large samples:

$$\hat{\theta}_x \sim N\left(\theta, \text{Var}\left(\hat{\theta}_X\right)\right), \quad \hat{\theta}_Y \sim N\left(\theta, \text{Var}\left(\hat{\theta}_Y\right)\right)$$

They only differ in the variance terms!

We say $\hat{\theta}_X$ is **more efficient** than $\hat{\theta}_Y$ if $\text{Var}\left(\hat{\theta}_X\right) < \text{Var}\left(\hat{\theta}_Y\right)$.

Both converge at the same $O(n^{-1/2})$ rate — efficiency is a competition over the **constant**. The rest of this video is about engineering τ down.

Antithetic Variables

Antithetic variables

So far, we have been generating **independent, identically distributed** collections of random variables.

With **antithetic variables**, we introduce **dependence** in a way that **decreases the variance** of the estimator.

The trick is to find a way to generate the dependence in a very particular way.

Thinking more about dependence and variance

Suppose we have an estimator for θ that is a sample mean of T_i: $\hat{\theta} = m^{-1} \sum_{i=1}^{m} T_i$.

The variance of this estimator is

$$\text{Var}(\hat{\theta}) = \frac{1}{m^2} \left[\sum_{i=1}^{m} \text{Var}\left(T_i\right) + \sum_{i \neq j} \text{Cov}(T_i, T_j) \right]$$

When T_i is independent of T_j the covariance term is zero.

What if we could generate **negatively correlated** T_i **and** T_j?

Example: Uniform random variables

If $U \sim U(0,1)$ then $U' = 1 - U$ has the **same distribution** as U, and

$$\text{Cov}(U, U') = E(U(1-U)) - (1/4) = E(U) - E(U^2) - 1/4 = 1/4 - 1/3 = -1/12$$

More generally, if t is **monotonic**, then $t(U)$ and $t(1-U)$ are also **negatively correlated**.

So: for the price of $m/2$ uniforms, we get m draws whose sample mean has *less* variance than m independent draws.

Example continued

Let's estimate

$$\theta = \int_0^1 u^{1/3}\, du = \mathsf{E}\left(U^{1/3}\right) = 3/4$$

(an integral we can check exactly). Note $t(u) = u^{1/3}$ is monotone.

To reduce the variance, we'll use pairs:

$$\hat{\theta} = \frac{1}{m/2} \sum_{i=1}^{m/2} \frac{t(U_i) + t(1 - U_i)}{2}$$

IID solution

```r
iids <- runif(10000)^(1/3)
mean(iids)
```

```
## [1] 0.7482
```

```r
(est_var_iid <- var(iids) / 10000)
```

```
## [1] 3.791e-06
```

To keep the sample size the same, let's only generate 5000 uniforms, then supplement those with 5000 copies of $1 - U$.

```r
tmp <- runif(5000)
antis <- (tmp^(1/3) + (1 - tmp)^(1/3)) / 2
mean(antis)
```

```
## [1] 0.7526
```

```r
(est_var_anti <- var(antis) / 5000)
```

```
## [1] 4.634e-07
```

Percent variance reduction

```r
1 - (est_var_anti / est_var_iid)
```

```
## [1] 0.8778
```

More than a 80% reduction in variance (and we only had to generate 1/2 as many random variables).

Note: it is not always obvious how to generate the antithetic variable pairs, but when you can, they are a very powerful tool.

Importance Sampling

Changing the sampling distribution

Recall the **multiply and divide** trick from last video. Even when our target *is already* an expectation,

$$\theta = \mathsf{E}\left(h(X)\right) = \int_{-\infty}^{\infty} h(x)f(x)\,dx$$

we can multiply and divide by another density:

$$\int_{-\infty}^{\infty} h(x)f(x)\,dx = \int_{-\infty}^{\infty} h(x)f(x)\frac{g(x)}{g(x)}\,dx = \mathsf{E}\left(h(Y)\frac{f(Y)}{g(Y)}\right)$$

for a random variable Y with density $g(y)$.

Why bother, when we could sample X directly? Because **a good choice of g can have far lower variance**.

Example: Tail probabilities

Sampling from X directly is a poor way to estimate **tail probabilities**

$$P(X \geq x)$$

since we get relatively few random samples from the tail.

Ex.: For $Z \sim N(0,1)$, what is $P(Z \geq 4.5) = E(\mathbf{1}(Z \geq 4.5))$?

```r
k <- 100000
sum(rnorm(k) >= 4.5)
```

```
## [1] 0
```

One hundred thousand draws and **not a single one** landed in the region we care about.

Using a shifted exponential

Consider instead drawing from $Y = 4.5 + \text{Exp}(1)$ so that

$$g(y) = \exp(-(y - 4.5)), \quad y > 4.5$$

$$\mathrm{E}\left(\mathbf{1}(Z \geq 4.5)\right) = \mathrm{E}\left(\frac{\mathbf{1}(Y \geq 4.5)\phi(Y)}{g(Y)}\right) = \mathrm{E}\left(\frac{\phi(Y)}{g(Y)}\right)$$

```r
ys <- rexp(k) + 4.5
ratios <- dnorm(ys) / (dexp(ys - 4.5))
mean(ratios)
```

```
## [1] 3.381e-06
```

```r
(truep <- pnorm(4.5, lower.tail = FALSE))
```

```
## [1] 3.398e-06
```

Variances

Recall, we prefer estimators that are **efficient** (have lower variance).

Variance of the direct (indicator) estimator (true, not estimated):

```
truep * (1 - truep) / k
```

```
## [1] 3.398e-11
```

Estimated variance of the shifted exponential version:

```
var(ratios) / k
```

```
## [1] 1.938e-16
```

Every sample from Y lands in the tail and carries a small, smooth weight — instead of almost every sample contributing exactly zero.

Terminology

We call using $\mathrm{E}\left(h(Y)f(Y)/g(Y)\right)$ to estimate $\mathrm{E}\left(h(X)\right)$ **importance sampling**.

- We call the distribution of Y the "envelope."
- The density of Y, $g(y)$, is the "importance function."
- We call the ratio $f(Y)/g(Y)$ the "importance weights."

The name: g concentrates our samples where they are **important** for the integral, and the weights undo the distortion.

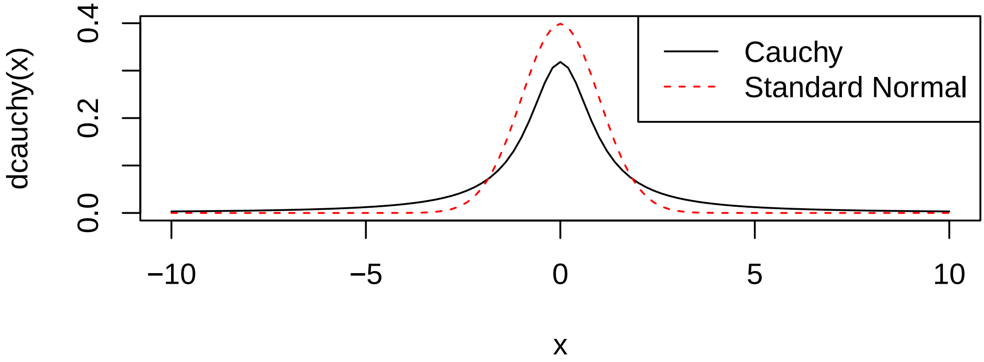
Picking the envelope distribution

The envelope must be **non-zero wherever f is** (we cannot weight samples we never see), and we can get into trouble when the ratio $f(x)/g(x)$ gets very large (e.g., f has "fatter tails" than g).

In particular, we need the importance sampling estimator to have finite variance for the law of large numbers and the CLT to hold:

$$E_Y\left(h(Y)^2\frac{f(Y)^2}{g(Y)^2}\right) = \int_{-\infty}^{\infty} h(y)^2\frac{f(y)^2}{g(y)^2}g(y)\,dy = E_X\left(h(X)^2\frac{f(X)}{g(X)}\right) < \infty$$

Example: Targeting the Cauchy distribution with the Normal

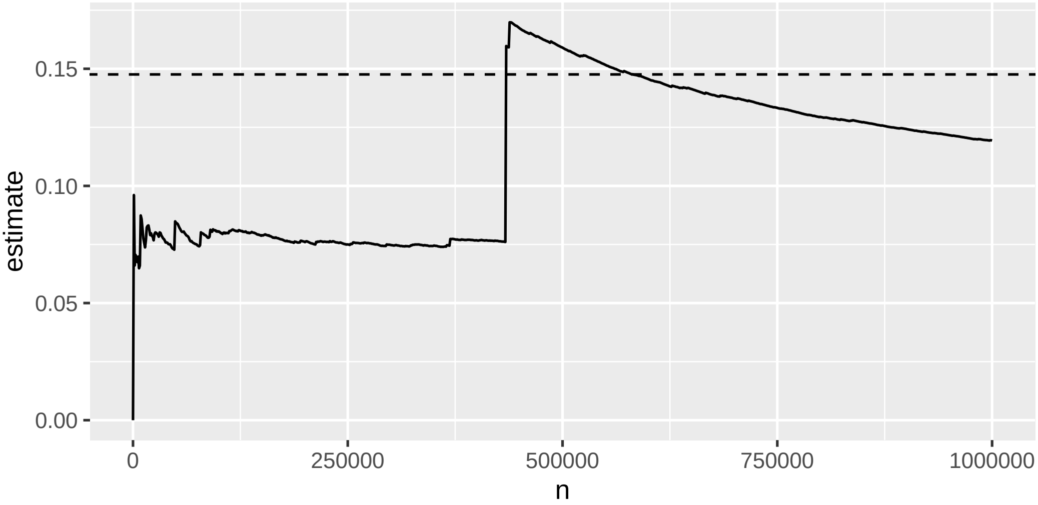


Cauchy from Normal

Let $C \sim \text{Cauchy}(0)$. Let's estimate $P(C \geq 2)$:

```
k <- 1000000 # one million samples
ys <- rnorm(k)
iweights <- dcauchy(ys) / dnorm(ys)
estimates <- cumsum(iweights * (ys >= 2)) / (1:k)
```

Plotting estimate vs. number of samples



Avoiding degenerate envelopes

Importance sampling for the Cauchy distribution will always be difficult due to the "fat tails."

Specifically, if the quantity

$$h(x)\frac{f(x)}{g(x)} < c, \quad \text{for some } c$$

then you should be ok.

The cases we'll consider in this class will be safe, but keep this in mind when using importance sampling in new ways.

Variance terms

Which envelopes are *good*? Recall that for **IID data**, the variance of the sample mean is

$$\mathrm{Var}\left(\hat{\theta}\right) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} h(Y_i)\frac{f(Y_i)}{g(Y_i)}\right) = \frac{1}{n}\mathrm{Var}\left(h(Y)\frac{f(Y)}{g(Y)}\right)$$

$$\begin{aligned}\mathrm{Var}\left(h(Y)\frac{f(Y)}{g(Y)}\right) &= \mathrm{E}\left(h(Y)^2\frac{f(Y)^2}{g(Y)^2}\right) - \left[\mathrm{E}\left(h(Y)\frac{f(Y)}{g(Y)}\right)\right]^2 \\ &= \int h(y)^2\frac{f(y)^2}{g(y)^2}\, g(y)\, dy - \theta^2\end{aligned}$$

Further decomposing variance

Notice that for any function $t(x)^2 = |t(x)|\,|t(x)|$.

$$\begin{aligned}
\mathrm{Var}\left(h(Y)\frac{f(Y)}{g(Y)}\right) &= \int h(y)^2 \frac{f(y)^2}{g(y)^2}\, g(y)\, dy - \theta^2 \\
&= \int |h(y)||f(y)| \frac{|h(y)||f(y)|}{g(y)}\, dy - \theta^2 \\
&= \int |h(y)|f(y) \frac{|h(y)|f(y)}{g(y)}\, dy - \theta^2
\end{aligned}$$

Picking a $g(y)$ to minimize variance

Observe that because $|h(x)|f(x) > 0$, the following is a proper density:

$$g(y) = \frac{|h(y)|f(y)}{\int_{-\infty}^{\infty} |h(y)|f(y)\,dy}$$

$$\begin{aligned}
\text{Var}\left(h(Y)\frac{f(Y)}{g(Y)}\right) &= \int |h(y)|f(y)\frac{|h(y)|f(y)}{g(y)}\,dy - \theta^2 \\
&= \int |h(y)|f(y)\left[\int |h(y)|f(y)\,dy\right]\,dy - \theta^2 \\
&= \left[\int |h(y)|f(y)\,dy\right]^2 - \theta^2
\end{aligned}$$

and if $h(y) > 0$ for all y where $f(y) > 0$, then the **variance would be zero!**

Using minimum variance g

So we should pick

$$g(y) = \frac{|h(y)|f(y)}{\int_{-\infty}^{\infty} |h(y)|f(y)\,dy}$$

Why is this not helpful? We would need to know $\int |h(x)|f(x)\,dx$, effectively θ.

Why is this helpful? We can try to pick $g(y) \approx c\,|h(y)|f(y)$.

Example: Truncated Normal

Consider the **truncated Normal distribution** $Z \mid 0 \leq Z \leq 1$, $Z \sim N(0,1)$, with density

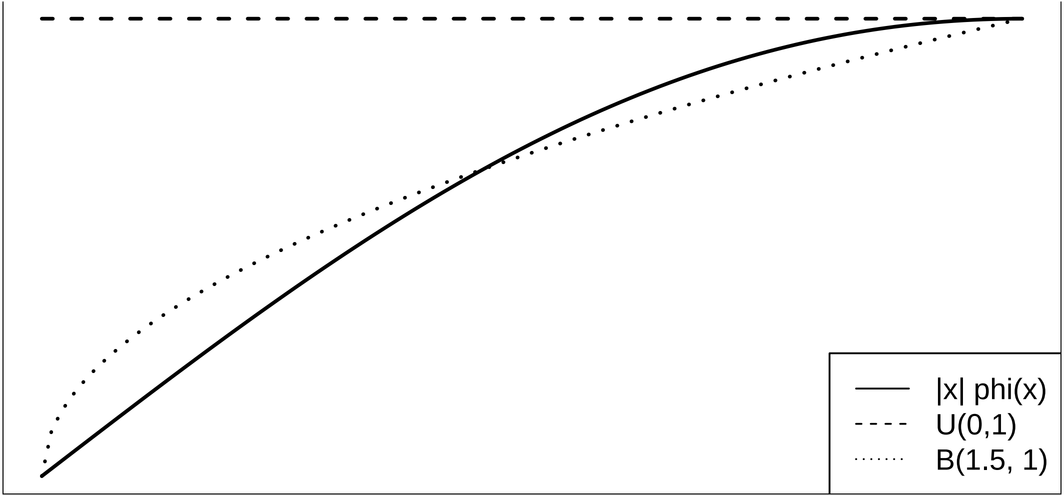
$$f(x) = \frac{\phi(x)}{\Phi(1) - \Phi(0)}, \quad 0 \leq x \leq 1$$

We want to estimate $\mathrm{E}(X)$, so $h(x) = x$ and the ideal envelope is $\propto |x|\phi(x)$: **small near 0, rising toward 1**.

Two built-in candidates on $(0,1)$:

- Uniform(0,1) — flat, ignores the shape
- Beta(1.5, 1) — density $1.5\sqrt{x}$, rising like the target

Graphing $|h(x)|f(x)$ against the candidates



Comparing the two envelopes

```
tn <- function(x) { dnorm(x) / (pnorm(1) - pnorm(0)) }
k <- 10000
yu <- runif(k)
iwu <- tn(yu) / dunif(yu)    ## f(y) / g(y)
mean(yu * iwu)
```

```
## [1] 0.4579
```

```
yr <- rbeta(k, 1.5, 1)
iwr <- tn(yr) / dbeta(yr, 1.5, 1)
mean(yr * iwr)
```

```
## [1] 0.4587
```

Comparing variance

```r
varu <- var(yu * iwu) # variance of single h(Y) f(Y) / g(Y) term
varr <- var(yr * iwr)
(varu - varr) / varu  # percent decrease, uniform -> Beta
```

```
## [1] 0.8775
```

Relative CI width:

```r
diff(t.test(yr * iwr)$conf.int) / diff(t.test(yu * iwu)$conf.int)
```

```
## [1] 0.35
```

The envelope that **tracks** $|h(x)|f(x)$ wins.

Further Reading in Chapter of SCR

Every estimator today was a **sample mean**, which is why the rate never budged and all the work went into τ. Two more techniques from the reading:

- **Control variates** (SCR §6.5–6.7): given a correlated $v(X)$ whose mean μ you know *exactly*, $h(X) + \beta(v(X) - \mu)$ is unbiased for **every** β — so choose the β with the least variance. Antithetic variables are the case $v(U) = t(1 - U) - t(U)$, $\mu = 0$, where the best β turns out to be $1/2$ for every t.
- **Stratified sampling** (SCR §6.7–6.8): partition the domain, sample each region separately, recombine. The cheapest of these to write, and on today's $u^{1/3}$ integral the most effective — 10 equal strata cut the variance 97%, against antithetic's 86%. But it needs k strata *per dimension*, so it fades in high dimensions for exactly the reason grid search did.

Summary

- The $O(n^{-1/2})$ rate is fixed: a $10\times$ better answer costs $100\times$ the draws. Variance reduction shrinks the **constant** τ instead.
- Two estimators of the same θ differ only in variance; the smaller is **more efficient**.
- **Antithetic variables**: pair each draw with a negatively correlated partner. Needs t monotone; here it cut the variance 86% using half as many uniforms.
- **Importance sampling**: sample from an envelope g, reweight by f/g. The ideal g is proportional to $|h|f$ — we cannot normalize that, but we can imitate it.
- It is the tool of choice for **tail probabilities**, where direct sampling puts almost no draws where they matter. Watch the weights: g with thinner tails than f blows up the variance.
- Importance sampling returns in Module 2: generating from distributions R doesn't have, and resampling with the weights.